

ASP .Net Core Web API-Part 1

Distributed Application Development in the
Cloud - NET 031

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Agenda

- HTTP
- Specification Verbs
- Status Codes
- Headers

HTTP Basics

- The Hypertext Transfer Protocol (HTTP) is an application layer protocol
- HTTP is the foundation of data communication for the World Wide Web
- All pages in web sites in the World Wide Web (WWW) are regarded as resources
- A resource is any piece of information that can be unambiguously identified by a Uniform Resource Identifier (URI)
- A URI consists of a URI scheme name such as “http” followed by a colon character and then by a scheme-specific part

HTTP Basics Features

- Connectionless
- Media Independent
- Stateless

HTTP Request

Each HTTP request contains encoded data, with information such as:

- A URL. This points to the resource on the web.
- An HTTP method. This indicates the specific action the request expects to receive from the server in its response.
- HTTP request headers
- An HTTP body

HTTP Methods (Verbs)

HTTP defines methods (verbs) to indicate the desired action to be performed on the identified resource.

GET

Gets a resource

POST

Used to insert a new item in a collection

PUT

Used to update an item in a collection

DELETE

Deletes an item in a collection

HTTP Status Codes

1xx

Informational

2xx

Successful

3xx

Redirection

4xx

Client Error

5xx

Server Error

Common HTTP Status Codes

- 200 - Ok
- 201 - Created
- 204 – No Content
- 301 – Moved Permanently
- 304- Not Modified
- 400 – Bad Request
- 401 – Unauthorized
- 403 – Forbidden
- 404 – Not Found
- 408 – Request Timeout
- 409 – Conflict
- 410- Gone
- 415- Unsupported Media Type

Common HTTP Status Codes

500 – Internal Server Error

502- Bad Gateway

503 – Service Unavailable

504 – Gateway Timeout

https://developer.mozilla.org/en-US/docs/Web/HTTP/Status#successful_responses

HTTP Header Fields

HTTP headers let the client and the server pass additional information with an HTTP request or response.

Accept

Content-Types that are acceptable for the response

Authorization

Authentication credentials for HTTP authentication

Content-Type

The MIME type of the body of the request (used with POST and PUT requests)

Content-Length

The length of the request body in octets (8-bit bytes)

HTTP Header Fields (cont)

Accept-Language

Accept-Language: en-us

Masked

True/False

Cache-Control

Cache-Control: max-age=3600

Retry-After

Retry-After: 120

Internet Media Types

- Internet media types were originally called MIME types (Multipurpose Internet Mail Extensions)
 - Their use has expanded to HTTP and other Internet protocols
- Used in the Accept HTTP Header or Content Type
 - Accept: application/xml
 - Content-Type: application/json

THANK YOU.



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